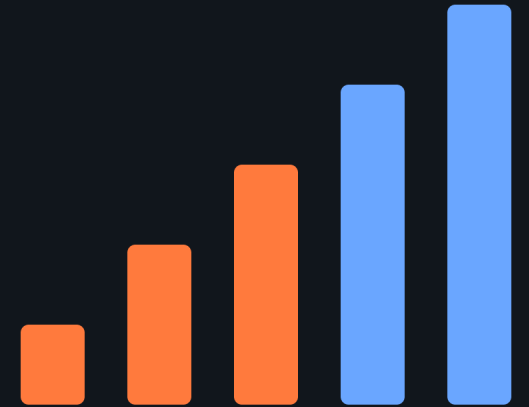


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Scale Software Development with AI Sandboxes

From laptop steering to agent swarms

Mathew Mathew · CTO / Claramap.com]





LET'S CONNECT

Mathew Mathew

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Claramap helps companies design, build and run **agentic systems** in production.

20+ years as a software engineer and data scientist at IBM Cloud, Amdocs, AT&T, SolarWinds and Discover.



SCAN TO CONNECT

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github.com/mathaix/fife

github.com/mathaix/signals

BEFORE WE START

Where are you on the ladder?

The diagram illustrates five levels of AI agent abstraction, arranged vertically. A central navigation bar is positioned between the second and third levels. The levels are numbered 1 to 5 from bottom to top. Each level includes a number, a title, a description, and a key feature or bottleneck.

Level	Agent Type	Description	Key Feature / Bottleneck
5	Agent swarm	Many agents, one problem	Isolation vs. speed vs. cost
4	Agent factory	Tickets in, PRs out	Review & verification
▲ SANDBOXES · · · · · LAPTOP ▼ BOTTLENECK			
3	Parallel local agents	Worktrees and YOLO mode	Your hardware
2	Local agent	An agent in your terminal	Your approvals
1	Autocomplete	AI suggests, you do the rest	Your typing

BOTTLENECK

What a sandbox is

An ephemeral, isolated computer with an API, where an agent can run anything inside a blast radius **you control.**

Isolated

Ephemeral

Programmable

Fast to start

Snapshottable

The isolation spectrum

OS sandbox

Seatbelt,
bubblewrap,
Landlock

Container

Namespaces and
cgroups; shared
host kernel

gVisor

User-space kernel
intercepts syscalls

microVM

Firecracker, Kata;
own kernel

Full VM

Full hardware
virtualization

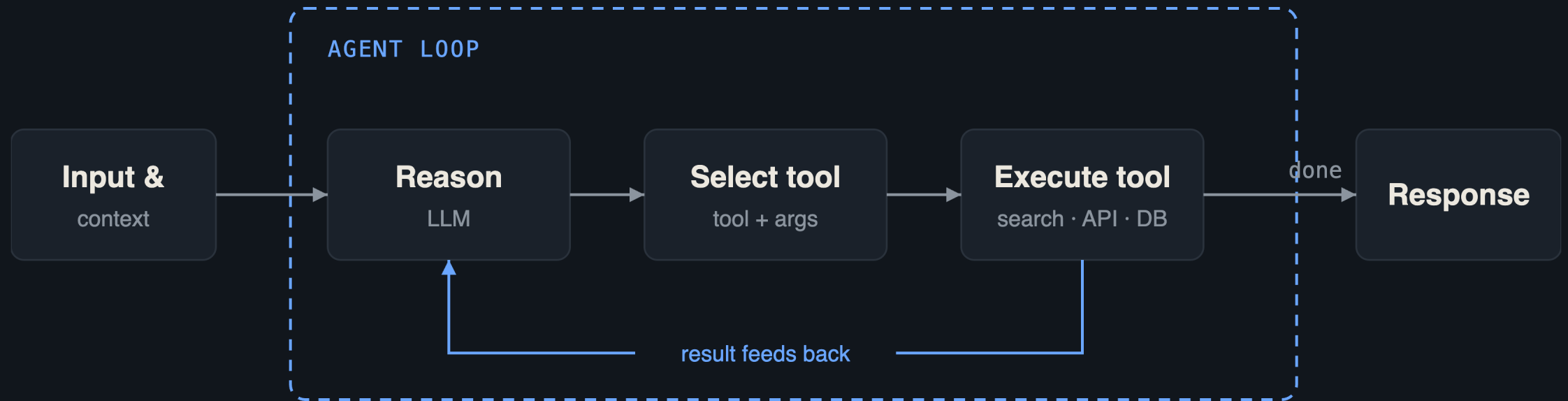
← FASTER START • LESS OVERHEAD

STRONGER BOUNDARY →

Who builds the sandboxes

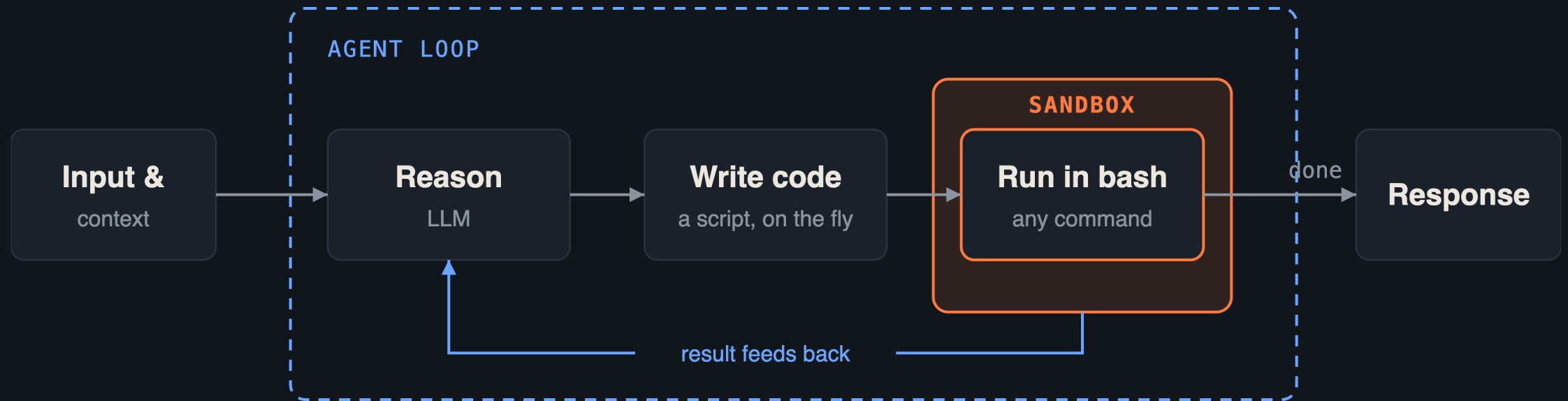
Tool	Isolation	Notable
Firecracker	microVM	Open-source building block from AWS; not a product
Docker	Container	Dev default; not enough alone for untrusted code
E2B	Firecracker microVM	Agent-focused code-interpreter SDK
Modal	gVisor	Built for scale; locked down by default
Daytona	Container (Sysbox)	Fast creation; egress allowlists; placeholder secrets

The agent loop



Every action is a tool you defined. You know exactly what it can do.

The coding agent



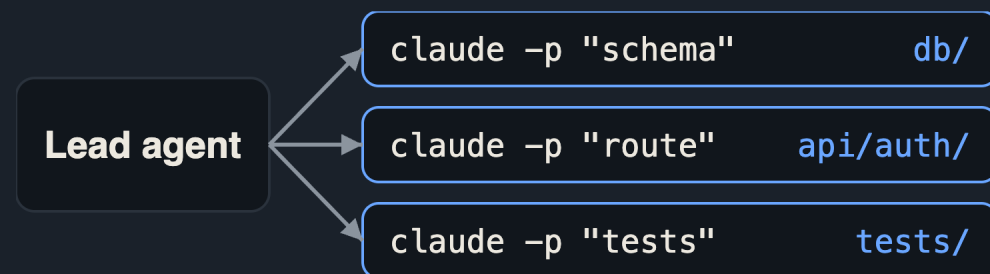
Give it bash and it writes its own tools. You can't predict what it will run, so run it in a sandbox.

Headless execution

```
$ claude -p "Fix the failing test in auth/" --output-format json
```

Decompose

Split a big task. Each subagent gets one job and only the context it needs.

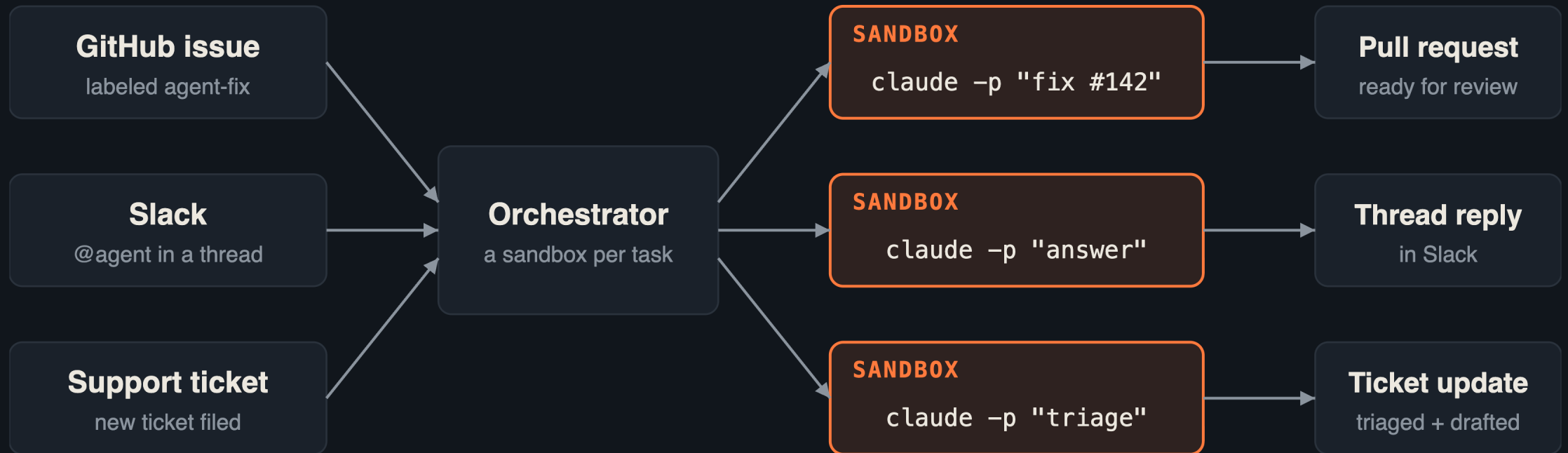


Automate

No one at the keyboard. Events and schedules start the agent.



Orchestrate



Same agent, many entry points. Every task gets its own sandbox.

Laptops are where agents learn to code.
Sandboxes are where they learn to scale.

Thank you · Questions?

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